10/3/22 Connor Madden

Extra Exercises 3

1) a) $p_1 = 5$ bar

$$V_1 = V_f + V_g$$

$$m = \frac{V_f}{v_f} + \frac{V_g}{v_g} = 9.16\text{ kg} = m$$

$$V_1 = V_2 \therefore V_2 = 1.505\ m^3$$

$$\therefore v_2 = \frac{1.505\ m^3}{9.16\text{ kg}} = .1643\ m^3/kg$$

$$p_2 = \frac{-(7-8)(.1815-.1643)}{(.1815-.1596)} + 7 = \boxed{7.783\text{ bar} = P_2}$$

b) $W = \Delta U - Q$, well insulated $\therefore Q = 0$ so $W = \Delta U$

$$W = m(u_2 - u_1)$$

$$u_2 = \frac{\cancel{-}(1328.04 - 1330.64)(7 - 7.783)}{(7 - 8)} + 1328.04$$

$$x_1 = \frac{m_g}{m} = \frac{6}{9.16} = .655$$

$$u_2 = 1330.08\text{ kJ/kg}$$

$$u_1 = u_f + x_1(u_g - u_f) = 198.39 + .655(1321.02 - 198.39)$$

$$u_1 = 933.94\text{ kJ/kg}$$

$$W = 9.16(1330.08 - 933.94)$$

$$\boxed{W = 3627.5\text{ kJ}}$$

2) a) $W = \frac{-(P_2V_2 - P_1V_1)}{1-n}$

100 kPa = 1 bar & −10°C

$$\therefore v_1 = 1.2621\ m^3/kg$$

$$18\text{ kg} \therefore V_1 = 22.7178\ m^3$$

$$P_2V_2^n = P_1V_1^n$$

$$V_2 = \left(\frac{P_1V_1^n}{P_2}\right)^{\frac{1}{n}} = 4.8229\ m^3$$

$P_1 = 100$ kPa $V_1 = 22.7178\ m^3$

$P_2 = 550$ kPa $V_2 = 4.8229\ m^3$

$$W = \frac{-((550)(4.8229) - (100)(22.7178))}{1 - 1.1}$$

$$\boxed{W = 3808.21\text{ kJ}}$$

550 kPa = 5.5 bar

b) $Q = \Delta U - W$

$$u_1 = 1324.33$$

$$Q = m(u_2 - u_1) - W$$

$$v_2 = \frac{4.8229}{18} = .26794$$

$$= 18(1396.22 - 1324.33) - 3808.21$$

$$1396.22\text{ kJ/kg} = u_2 = \frac{-(1[illegible]9.84 - 14[illegible].53)(.26441 - .26794)}{(.26441 - .27454)} + 13[illegible]$$

$$\boxed{Q = -2514.16\text{ kJ}}$$